

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Single Correct

Given below are two statements:

Statement I : Both $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ and $\text{MgCl}_2 \cdot 8\text{H}_2\text{O}$

undergo dehydration on heating. Statement II : BeO is amphoteric whereas the

oxides of other elements in the same group are

acidic.

In the light of the above statements, choose the

correct answer from the options given below :

(1) Statement I is false but statement II is true

(2) Both statement I and statement II are false

(3) Both statement I and statement II are true

(4) Statement I is true but statement II is false

Q2: 16 March (Shift 2) - Single Correct

Statement I : Sodium hydride can be used as an oxidising agent.

Statement II : The lone pair of electrons on nitrogen in pyridine makes it basic.

Choose the CORRECT answer from the options given below:

(1) Both statement I and statement II are false

(2) Statement I is true but statement II is false

(3) Statement I is false but statement II is true

(4) Both statement I and statement II are true

Q3: 17 March (Shift 1) - Single Correct

The correct order of conductivity of ions in water is :

(1) $\text{Na}^+ > \text{K}^+ > \text{Rb}^+ > \text{Cs}^+$

(2) $\text{Cs}^+ > \text{Rb}^+ > \text{K}^+ > \text{Na}^+$

(3) $\text{K}^+ > \text{Na}^+ > \text{Cs}^+ > \text{Rb}^+$

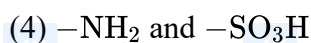
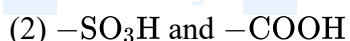
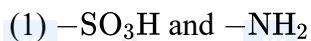
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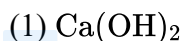
Q4: 17 March (Shift 2) - Single Correct

The functional groups that are responsible for the ion-exchange property of cation and anion exchange resins, respectively, are :



Q5: 17 March (Shift 2) - Single Correct

One of the by-products formed during the recovery of NH_3 from Solvay process is :

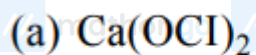


Q6: 18 March (Shift 1) - Single Correct

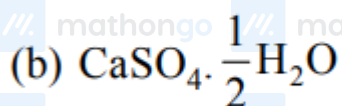
Match List-I with List-II

List-I

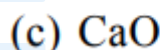
List-II



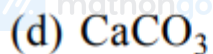
(i) Antacid



(ii) Cement



(iii) Bleach



(iv) Plaster of paris

Questions with Answer Keys

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Choose the most appropriate answer from the options given below:

(1) a – i, b – iv, c – iii, d – ii

(2) a-iii, b-ii, c – iv, d – i

(3) a-iii, b-iv, c – ii, d – i

(4) a-iii, b – ii, c – i, d – iv

Q7: 18 March (Shift 1) - Single Correct

Given below are two statements: One is

labelled as Assertion A and the other labelled

as reason R

Assertion A : During the boiling of water having temporary hardness, $\text{Mg}(\text{HCO}_3)_2$ is converted to MgCO_3

Reason R : The solubility product of $\text{Mg}(\text{OH})_2$ is greater than that of MgCO_3 . In the light of the above statements, choose the most appropriate answer from the options given

below:

(1) Both A and R are true but R is not the correct explanation of A

(2) A is true but R is false

(3) Both A and R are true and R is the correct explanation of A

(4) A is false and R is also false

Q8: 18 March (Shift 2) - Single Correct

The secondary valency and the number of hydrogen bonded water molecule(s) in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, respectively, are

(1) 6 and 4

(2) 4 and 1

(3) 6 and 5

(4) 5 and 1

Q9: 18 March (Shift 2) - Single Correct

Match list-I with list-II :

List-I**List-II**

(a) Be

(i) Treatment of cancer

(b) Mg

(ii) Extraction of metals

(c) Ca

(iii) Incendiary bombs and signals

(d) Ra

(iv) Windows of X-ray tubes

(v) Bearings for motor engines.

Choose the most appropriate answer the option given below:

(1) a – iv, b – iii, c – i, d – ii

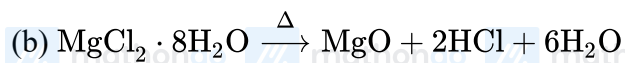
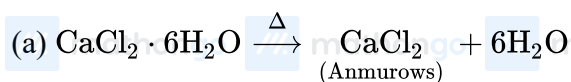
(2) a – iv, b – iii, c – ii, d – i

(3) a – iii, b – iv, c – v, d – ii

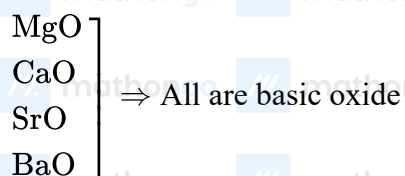
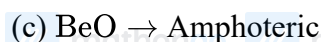
(4) a – iii, b – iv, c – ii, d – v

Answer Key**Q1 (2)****Q2 (3)****Q3 (2)****Q4 (1)****Q5 (3)****Q6 (3)****Q7 (4)****Q8 (2)****Q9 (2)**

Q1 (2)



The dehydration of hydrated chloride of calcium can be achieved. The corresponding hydrated chloride of magnesium on heating suffer hydrolysis.



Q2 (3)

(1) NaH (sodium Hydride) is used as a reducing reagent.



(2)

In pyridine, due to free electron on N atom, it is basic in nature.

Hence statement I is false & II is true.

Q3 (2)

As the size of gaseous ion decreases, it get more hydrated in water and hence, the size of aqueous ion increases. When this bulky ion move in solution, it experience greater resistance and hence lower conductivity.

Size of gaseous ion : $\text{Cs}^+ > \text{Rb}^+ > \text{K}^+ > \text{Na}^+$

Size of aqueous ion : $\text{Cs}^+ < \text{Rb}^+ < \text{K}^+ < \text{Na}^+$

Conductivity : $\text{Cs}^+ > \text{Rb}^+ > \text{K}^+ > \text{Na}^+$

Q4 (1)

Cation exchanger contains $-\text{SO}_3\text{H}$ or $-\text{COOH}$ groups while anion exchanger contains basic groups like $-\text{NH}_2$

Q5 (3)

Hints and Solutions

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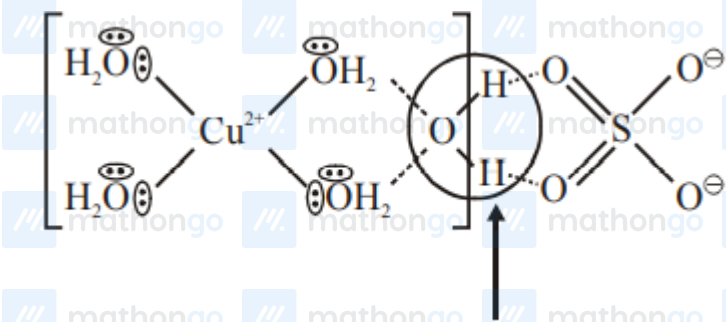
Q6 (3)

 $\text{Ca}(\text{OCl})_2$ is Bleach. $\text{CaSO}_4 \cdot \frac{1}{2} \text{H}_2\text{O}$ is plaster of paris. CaCO_3 is used as an antacid. CaO is major component of cement.

Q7 (4)

For temporary hardness, $\text{Mg}(\text{HCO}_3)_2 \xrightarrow{\text{heating}} \text{Mg}(\text{OH})_2 \downarrow + 2\text{CO}_2 \uparrow$ Assertion is false. MgCO_3 has high solubility product than $\text{Mg}(\text{OH})_2$ According to data of NCERT table 7.9 (Equilibrium chapter), the solubility product of magnesium carbonate is 3.5×10^{-8} and solubility product of $\text{Mg}(\text{OH})_2$ is 1.8×10^{-11} Hence Reason is incorrect.

Q8 (2)



Hydrogen bonded water molecule = 1 Secondary valency = 4

Q9 (2)

(a) Be $\rightarrow$ it is used in the Windows of X-ray tubes(b) Mg $\rightarrow$ it is used in the Incendiary bombs and signals(c) Ca $\rightarrow$ it is used in the Extraction of metals(d) Ra $\rightarrow$ it is used in the Treatment of cancer